NORMATIVE ARCHITECTURE SPECIFICATION

GibiWorld

Mobile AR pet companion, spatial game platform, and secure Pawsome3D asset runtime

Implementation baseline for client, backend, asset, QA, security, live operations, and data teams

Version 1.0 | 31 July 2026

Document status: APPROVED DESIGN BASELINE. Every statement containing SHALL, MUST, MUST NOT, REQUIRED, or EXACTLY is normative. Deviations require a written Architecture Decision Record and a new document version.

Document control

Field	Value
Specification	GW-ARCH-001
Version	1.0.0
Baseline date	2026-07-31
Client	Unity mobile AR for iOS and Android
Spatial baseline	Niantic Spatial SDK 4.1.0, VPS2, AR Foundation providers
Asset authority	Pawsome3D or GibiWorld preset catalog only
Change control	Pull request + ADR + compatibility and security review

0. Executive decisions

- The shipping client SHALL pin Unity 6000.0.74f1, Niantic Spatial SDK 4.1.0, and AR Foundation 6.4.1. Package versions SHALL be frozen in packages-lock.json. An upgrade is a release project, not an automatic dependency update.
- The game SHALL use VPS2 map-relative anchors for persistent, shared, and ranked content. It SHALL use a local plane or live-mesh anchor for private unranked play when no processed VPS site is available.
- A pet SHALL render only if its immutable manifest names issuer PAWSOME3D or GIBIWORLD_PRESET, the Ed25519 signature verifies against a pinned trusted key, the SHA-256 digest matches the downloaded GLB, and the authenticated user has a current entitlement.
- Runtime pet payloads SHALL be GLB 2.0 loaded through glTFast. GibiWorld SHALL never execute scripts, shaders, animation events, or arbitrary code supplied by a pet asset.
- Frame-level behavior, physics, navigation, course scoring, and safety SHALL be deterministic on the client or authoritative session server. AI MAY choose only validated high-level intents from a fixed allowlist.
- The Pawsome3D integration SHALL use an API and signed webhook adapter. GibiWorld SHALL NOT read or write the Pawsome3D production database directly.
- Ranked scoring SHALL pause when the tracked site anchor is not in tracked state. Device VPS2 state alone SHALL NOT authorize placement or scoring.
- Launch accounts SHALL be age 13 or older. Under-13 accounts SHALL remain disabled until verifiable parental consent, guardian controls, child privacy review, and store declarations are implemented.

1. Product scope

1.1 In scope

- Personal companion play with a Pawsome3D or preset dog, cat, or approved domestic companion species.
- Stable local and VPS2 placement, environmental occlusion, live mesh collision, lighting adaptation, foot IK, gaze, spatial audio, and haptics.
- Kind training, fetch, tricks, exploration, course building, agility runs, photo moments, scrapbook memories, and optional shared sessions.
- Backend-owned profile, entitlement, bond, inventory, course, leaderboard, moderation, AI intent, telemetry, and live-configuration services.
- Security, privacy, accessibility, operational SLOs, test gates, and phased launch controls.

1.2 Explicitly out of scope

- Arbitrary model uploads, marketplace models, NFTs, user-supplied shaders, user scripts, or direct URLs.
- Wild animals, humans, fantasy combat creatures, or species not present in the server allowlist.
- Combat, pet injury, punishment, hunger loss, death, gambling, paid random rewards, or coercive daily streaks.
- Autonomous AI control of locomotion, collision, transactions, moderation, scoring, or safety decisions.
- World-scale promise of precise anchoring in areas without a processed VPS Site.

2. Experience pillars

Pillar	Implementation consequence	Acceptance signal
Present	Correct scale, ground contact, occlusion, light response, spatial audio, gaze, and animation continuity.	Users can walk around the pet without visible floor sliding or anchor jumps.
Personal	Only owned/selected pets, durable bond state, consented game memories, named rituals, and a private scrapbook.	A returning pet recalls approved preferences without inventing personal facts.
Kind	No punishment, deprivation, injury, or anxiety mechanics; rest and offline time never reduce bond.	Every training loop can end positively and no dark pattern penalizes absence.
Spatial	The real surface, safe volume, semantic tags, and anchor quality determine what can happen.	Courses fail closed when geometry or anchor quality is unsafe.
Trustworthy	Signed assets, server-authoritative value, privacy minimization, transparent status, and recoverable failure states.	Tampered files, stale entitlements, and untracked ranked runs are rejected.

3. System context and deployment

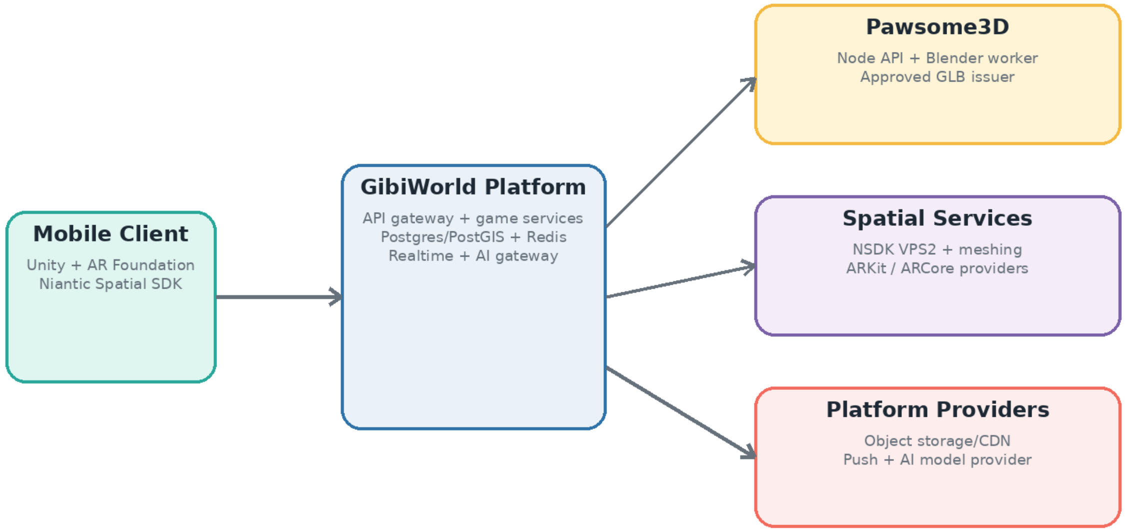

Figure 1. GibiWorld system context and trust boundaries

The platform is split into a frame-critical mobile runtime, stateless application services, data services, and provider adapters. The client never receives infrastructure credentials. Every external dependency is accessed through a named adapter so provider changes do not alter domain models.

3.1 Deployable units

Unit	Type	Responsibility	Scale
gw-mobile	Unity IL2CPP app	AR session, rendering, local simulation, cache, input, safety UI	iOS / Android
gw-edge-api	TypeScript service	TLS termination, JWT validation, rate limits, API routing	Stateless, multi-instance
gw-game-service	TypeScript service	Profile, pet state, inventory, bond, training, live config	Stateless, Postgres
gw-asset-service	TypeScript service	Issuer webhook, validation workflow, manifests, entitlement, revocation	Stateless + worker queue
gw-asset-worker	Blender + validator	Quarantine parse, canonicalization, LOD/texture/rig validation, thumbnails	Isolated worker, no public ingress
gw-world-service	TypeScript service	PostGIS queries, site/course versions, safe-zone metadata, nearby discovery	Stateless, PostGIS
gw-session-service	Realtime service	Presence, shared pet transforms, cooperative course sessions	WebSocket/QUIC capable
gw-ai-orchestrator	TypeScript service	Context assembly, model provider adapter, JSON schema validation, memory proposals	Stateless, strict egress
gw-moderation	Worker service	Names, course reports, venue reports, scrapbook share review	Queue-driven

Unit	Type	Responsibility	Scale
gw-telemetry	Collector	OpenTelemetry traces, metrics, structured events, privacy filters	Regional ingestion
gw-admin	Web console	Catalog, keys, revocation, course publication, feature flags, incidents	SSO + hardware-key MFA

3.2 Existing Pawsome3D compatibility

The known Pawsome3D environment uses a Node application and Blender worker on Render, SQL on Hostinger, a Backblaze B2 bucket, and a Hostinger-hosted website. It produces GLB pets from photo or text input. GibiWorld SHALL preserve that system as the asset authoring authority while creating a narrow integration boundary:

- Pawsome3D publishes an asset-ready webhook signed with HMAC-SHA256 and containing an immutable source asset ID, owner subject, source version, expiring server fetch URL, and generation timestamp.
- The GibiWorld asset service fetches the GLB server-side into quarantine. The mobile client never fetches a Pawsome3D production URL.
- Pawsome3D remains source of truth for creation and source ownership; GibiWorld remains source of truth for game compatibility, published digest, revocation, and in-game entitlement.
- Exact legacy endpoint names and database variable names SHALL be mapped in a deployment-only PawsomeAdapter configuration and SHALL NOT be compiled into the mobile app or this domain contract.

4. Unity client architecture

Assembly	Allowed dependencies	Responsibilities
Gibi.Core	None	Value objects, IDs, clocks, result types, deterministic math
Gibi.Spatial	Core, AR Foundation, NSDK adapter	Anchor quality, raycasts, planes, mesh, semantics, coordinate conversion
Gibi.Pets	Core, AssetRuntime, Animation	Pet entity, needs, bond presentation, local behavior state
Gibi.AssetRuntime	Core, glTFast	Manifest verification, download, cache, GLB instantiate, material policy
Gibi.Gameplay	Core, Spatial, Pets	Training, toys, courses, scoring client, safety gates
Gibi.Networking	Core	REST, WebSocket, token refresh, retries, idempotency, offline outbox
Gibi.AI	Core, Pets, Networking	Intent requests and validated response mapping; no model SDK in client
Gibi.UI	All public facades only	Screens, AR HUD, accessibility, localization, diagnostics
Gibi.Telemetry	Core	Consent-aware events, performance sampling, crash breadcrumbs
Gibi.Tests	All	EditMode, PlayMode, device simulation, recorded AR playback

Dependency rule: Assemblies SHALL reference inward through public interfaces. No assembly may call a provider SDK directly except its named adapter. Cyclic assembly references are a build failure.

4.1 Scene contract

- Bootstrap scene: dependency container, app lifecycle, authentication, remote config, crash handler, and persistent UI root only.
- ARWorld scene: exactly one ARSession and exactly one XR Origin (Mobile AR); ARCameraManager, AROcclusionManager, ARPlaneManager, ARRaycastManager, ARAnchorManager, ARMeshManager, semantic manager, and NSDK session adapter.
- PetSandbox scene: PetRoot, deterministic motion controller, animation graph, IK graph, interaction volumes, spatial audio emitter, and effects pool.
- Course scene content SHALL be data-driven and instantiated as pooled prefabs or signed GLB props from the GibiWorld catalog. A course payload shall never name a Resources path or execute an animation event.

4.2 Frame update order

1. Read platform input and AR provider frame.
2. Update NSDK/AR Foundation trackables and anchor quality snapshot.
3. Apply floating-origin and anchor transform corrections with smoothing limits.
4. Run deterministic pet perception and behavior intent arbitration at 10 Hz.
5. Run navigation/motion at FixedUpdate 50 Hz; never couple locomotion distance to render frame rate.
6. Evaluate Animator and Animation Rigging foot/gaze constraints.
7. Render URP camera, depth occlusion, transparent effects, UI, and accessibility overlays.
8. Sample telemetry after presentation; telemetry must never block the frame.

5. Spatial architecture

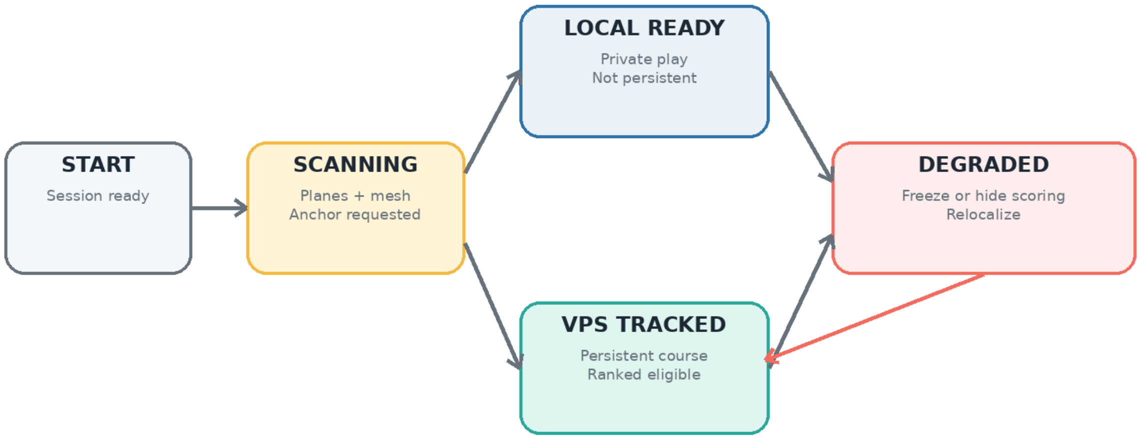

Figure 2. Placement eligibility and anchor degradation state machine

5.1 Coordinate frames

Frame	Representation	Authority	Use
WGS84	latitude/longitude/altitude float64	World service	Discovery, distance, site search, audit
VPS_MAP	right-handed map-relative pose from NSDK	VPS2 Site	Persistent/shared placement and ranked courses
AR_LOCAL	Unity float pose under XR Origin	Device AR session	Frame rendering and local physics
ANCHOR_LOCAL	position Vector3 + normalized quaternion	Course or object record	Child objects relative to one tracked anchor
PET_LOCAL	root-space meters	Pet motion controller	Animation, IK, interaction volumes

- Authoritative geographic values SHALL remain float64 on server and in transport. They SHALL NOT be stored in Unity Vector3.
- Persistent transforms SHALL be stored relative to one VPS Site anchor. World positions computed from a device session SHALL NOT be persisted.
- Quaternions SHALL be normalized to absolute error <= 1e-4 before persistence; zero or NaN quaternions are rejected.
- Course objects SHALL be within 75 meters of their site anchor. Larger experiences require multiple linked site segments.
- Floating-origin recentering SHALL occur only outside active scoring and SHALL preserve anchor-local transforms to <= 2 mm numerical error.

5.2 Anchor eligibility

State	Entry condition	Visible behavior	Scoring
UNAVAILABLE	No AR session or provider unavailable	2D companion card only	Disabled
SCANNING	Session tracking; no accepted surface	Scan coaching and mesh dots	Disabled
LOCAL_READY	Accepted plane/mesh anchor and clearance	Pet appears; LOCAL badge	Practice/ personal best only
VPS_LIMITED	Site anchor limited	Preview ghost only	Disabled
VPS_TRACKED	Target anchor state tracked for >= 1.0 s	Persistent content solid	Eligible
DEGRADED	Tracked lost > 250 ms or pose jump exceeds threshold	Freeze course clock; fade object; relocalization guide	Paused, then invalid after timeout

5.3 Spatial tagging contract

```
{
  "schemaVersion": 1,
  "spatialObjectId": "spo_01J...",
  "anchorFrame": "VPS_SITE",
  "siteId": "site_01J...",
  "anchorId": "vps-anchor-id",
  "pose": {"p": [1.250, 0.000, -3.800], "q": [0.0, 0.0, 0.0, 1.0]},
  "semanticTags": ["ground", "outdoor", "path"],
  "safety": {"clearanceRadiusM": 1.5, "clearanceHeightM": 2.0,
             "hazardClass": "NONE", "validatedAt": "2026-07-31T18:00:00Z"},
  "content": {"type": "AGILITY_GATE", "catalogId": "gate_basic_01", "version": 3}
}
```

- anchorFrame SHALL be LOCAL_SESSION or VPS_SITE. LOCAL_SESSION objects SHALL NOT be accepted by persistence endpoints.
- semanticTags SHALL come from the published enum. Unknown tags are retained only in authoring drafts and block publication.
- The runtime SHALL block sky, person, vehicle, water, road, rail, and unknown hazard regions from pet destination and obstacle placement.
- Surface acceptance SHALL require estimated slope <= 12 degrees for pet idle/training and <= 7 degrees for ranked gates, unless the obstacle profile explicitly permits a ramp.
- The user-facing placement ring SHALL encode status with color, icon, label, and optional haptic; color alone is insufficient.

6. Pawsome3D asset trust and publishing

Figure 3. Closed pet asset publishing and runtime verification pipeline

6.1 Canonical pet package

Artifact	Format	Required constraints
Model	glTF 2.0 Binary .glb	One scene; meters; +Y up; Unity-facing +Z after importer conversion; no external URI; no cameras/lights/scripts
Manifest	Canonical JSON UTF-8	RFC 8785 canonicalization; schema version; IDs; issuer; digest; rig; clips; bounds; size; compatibility
Signature	Ed25519 detached	Signature over canonical manifest bytes; key ID present; public key pinned and remotely revocable
Preview	WebP or PNG	Square 1024 px; opaque/transparent allowed; no executable metadata
Entitlement	Server record + short-lived JWT receipt	user subject, petAssetId, version, status, issuedAt, expiresAt
Provenance	JSON audit record	Pawsome source ID, worker build, validator version, source/output hashes, approvals

6.2 Geometry, material, texture, and rig limits

Dimension	Normative limit	Failure action
Species	dog, cat, rabbit, guinea_pig, ferret, miniature_pig; launch allowlist is server-controlled	Reject
Scale	shoulder height 0.12-1.10 m; total bounds <= 2.0 m on any axis	Reject
LOD0	<= 35,000 triangles, <= 45,000 vertices	Reject
LOD1	<= 18,000 triangles; screen transition 0.42	Worker may generate once; otherwise reject

Dimension	Normative limit	Failure action
LOD2	<= 7,500 triangles; screen transition 0.18	Worker may generate once; otherwise reject
LOD3	<= 2,000 triangles; screen transition 0.06	Worker may generate once; otherwise reject
Skinned meshes	<= 2; <= 4 weights/vertex; <= 96 deform bones; no scale animation	Reject
Materials	<= 3; PBR metallic-roughness only; alpha blend limited to fur cards/eyes	Reject
Textures	Base/normal <= 2048; other maps <= 1024; total decoded <= 48 MiB; KTX2 preferred	Transcode or reject
Morph targets	<= 12; <= 4 active simultaneously	Reject
GLB transfer size	<= 45 MiB compressed	Reject
Animation clips	<= 48; total keyframes <= 300,000; no animation events	Reject

6.3 Skeleton and animation profile

Playable pets SHALL use Unity Generic animation. The manifest skeletonProfile SHALL be one of GIBI_QUADRUPED_V1, GIBI_SMALL_MAMMAL_V1, or another versioned profile accepted by the server. Joint names are case-sensitive.

Group	Exact names	Rule
Root chain	root, pelvis, spine_01, spine_02, chest, neck, head	Required
Front left	clavicle_l, upper_front_l, lower_front_l, paw_front_l	Required
Front right	clavicle_r, upper_front_r, lower_front_r, paw_front_r	Required
Rear left	upper_rear_l, lower_rear_l, hock_l, paw_rear_l	Required
Rear right	upper_rear_r, lower_rear_r, hock_r, paw_rear_r	Required
Face	jaw, eye_l, eye_r	Required where anatomy has them
Tail	tail_01..tail_06	Optional, contiguous, max 6
Ears	ear_l_01..03, ear_r_01..03	Optional, max 3 each

Clip ID	Loop	Duration	Purpose
idle_a	loop	2-6 s	neutral standing
idle_b	loop	2-8 s	secondary quiet motion
walk	loop in-place	0.6-1.5 s	1.0 m/s reference
trot	loop in-place	0.5-1.2 s	2.0 m/s reference
run	loop in-place	0.4-1.0 s	3.8 m/s reference
turn_l_90 / turn_r_90	non-loop	0.25-0.8 s	root rotation disabled at import
sit / sit_idle / stand	mixed	0.4-4.0 s	training transitions
down / down_idle / rise	mixed	0.4-4.0 s	rest transitions

Clip ID	Loop	Duration	Purpose
jump_takeoff / jump_air / jump_land	mixed	0.15-1.0 s	course phases
pickup / carry / drop	mixed	0.2-2.0 s	mouth socket interaction
success / greet / pet_react / sleep	mixed	0.5-8.0 s	heartwarming behavior set

- All locomotion clips SHALL be in-place. The deterministic motion controller owns translation and yaw.
- Clip sampling SHALL be 30 fps or lower after key reduction. Scale curves are forbidden. Root transforms SHALL remain identity within 1e-4.
- Foot IK SHALL raycast against the accepted navigation surface or filtered mesh; max vertical correction 0.18 m, max paw rotation 25 degrees, blend in/out >= 100 ms.
- Look-at SHALL clamp head yaw to 50 degrees and pitch to 30 degrees; eye motion may add 15 degrees. Target changes SHALL use damped motion and never snap in one frame.
- Pet colliders SHALL be authored runtime primitives. Model mesh colliders are forbidden.

6.4 Runtime verification algorithm

1. Fetch manifest over TLS using authenticated asset endpoint. Reject unknown schema version, issuer, key ID, pet version, or compatibility range.
2. Canonicalize the manifest excluding signature and verify Ed25519 signature using the pinned key identified by keyId.
3. Confirm the authenticated entitlement status is ACTIVE and names the exact petAssetId and assetVersion.
4. Download to a temporary cache key. Stream SHA-256 during download; enforce Content-Length and 45 MiB hard limit.
5. Compare digest using constant-time equality. On mismatch, delete temporary bytes, emit asset_integrity_failed, and disable automatic retry for that URL.
6. Parse with glTFast using an import settings policy that disallows external URIs. Enforce node, mesh, material, texture, animation, and bounds limits again client-side.
7. Instantiate under PetAssetRoot, replace materials with the approved URP shader family, bind the Gibi controller, then atomically promote cache entry by digest.
8. Never persist signed URLs, access tokens, or raw authorization headers in logs, analytics, crash reports, or player-visible diagnostics.

7. Rendering, animation, and audio

Budget	Tier A/B target	Tier C fallback	Gate
Frame rate	60 fps; p95 CPU <= 16.7 ms; p95 GPU <= 16.7 ms	30 fps; p95 <= 33.3 ms	10-minute representative run
Memory	Peak resident <= 1.2 GiB	Peak resident <= 900 MiB	No OS memory warning

Budget	Tier A/B target	Tier C fallback	Gate
Pet rendering	<= 3 materials, <= 45k LOD0 vertices	Force LOD1, simplified fur	Frame debugger
World mesh	Visible collider mesh <= 120k triangles	<= 60k; 5 m cull	Runtime counter
Particles	<= 20k visible particles	<= 5k	Overdraw heat map
Draw calls	<= 160 in AR gameplay	<= 100	Frame debugger p95
App start	p95 <= 5 s to interactive home	p95 <= 7 s	Cold launch, cached auth
AR ready	p95 <= 4 s after camera permission	p95 <= 7 s	Supported lighting

- URP SHALL be used. The exact URP package version SHALL be the version resolved by the pinned Unity editor and recorded in packages-lock.json.
- Depth occlusion SHALL be enabled when supported. Without depth, the runtime SHALL use accepted plane/mesh occluders and visually reduce interactions that depend on hidden geometry.
- Real-time shadow quality SHALL be tiered: one main-light pet shadow, 1024 map maximum on Tier A, blob/contact shadow fallback on lower tiers.
- Dynamic resolution MAY vary 0.70-1.00. It SHALL not change UI resolution or break camera-to-depth alignment.
- Spatial audio SHALL use distance attenuation with a 0.5 m minimum and 8 m maximum for pet voice. Critical safety messages SHALL be non-spatial UI audio with captions and haptics.

8. Pet simulation and AI behavior

8.1 Deterministic behavior layers

Priority	Layer	Owner	Examples
0 highest	Safety override	Client deterministic	Stop, freeze, return to safe zone, cancel jump
1	Session rule	Authoritative course/session	Start, gate order, scoring pause, finish
2	Direct player cue	Client validated input	Sit, come, fetch, stay
3	AI high-level intent	Backend proposal, client validator	Greet, inspect toy, seek shade, invite play
4	Needs scheduler	Client deterministic	Rest, idle variety, look around
5	Ambient animation	Client	Blink, ear twitch, breathing

The arbiter evaluates at 10 Hz. A lower-priority intent cannot interrupt a locked safety, course, or player-cue action. Every accepted intent carries a monotonic sequence, start time, maximum duration, interruptibility, and fallback state.

8.2 AI intent contract

```
{
  "schemaVersion": 1,
  "requestId": "req_01J...",
  "petId": "pet_01J...",
  "contextRevision": 184,
  "intent": "INVITE_PLAY",
  "targetId": "toy_01J...",
  "emotion": {"valence": 0.25, "arousal": 0.15},
  "utteranceKey": "pet.invite_play.03",
  "memoryProposals": [],
  "expiresAt": "2026-07-31T18:00:10Z"
}
```

- Allowed intents SHALL be a server-published enum. Unknown intents, expired responses, wrong contextRevision, non-owned target IDs, or out-of-range emotion values are rejected.
- AI SHALL NOT output animation clip names, coordinates, physics forces, prices, inventory grants, leaderboard scores, moderation results, or executable text.
- The client SHALL map intent to a locally authored behavior. Missing mappings fall back to CALM_IDLE without error to the player.
- memoryProposals SHALL be server-validated and require an allowlisted fact type. Sensitive or inferred attributes SHALL be rejected. User deletion SHALL tombstone the fact and remove it from future context within 24 hours.
- The AI provider SHALL receive a rotating opaque player pseudonym, minimal approved memory, coarse time context, current pet state, and nearby game object types. It SHALL NOT receive camera frames, precise coordinates, contacts, advertising ID, or raw voice audio.

8.3 Offline behavior

The app SHALL contain a complete local behavior library sufficient for placement, idle, direct cues, training, fetch, and unranked courses. AI outage is a feature degradation, not a gameplay outage. The UI SHALL not display an error unless the player explicitly opens online memory or dialogue features.

9. Gameplay mechanics

9.1 Training state machine

State	Entry	Exit	Timeout
READY	Stable safe surface and pet available	Player starts lesson	None
CUE	Lesson active	Recognized cue or cancel	10 s -> gentle prompt
ACTION	Validated cue	Animation/action complete	Clip + 1 s -> abort safely
REWARD	Successful action	Player rewards or auto-kind close	8 s
REST	3 attempts or low energy	End or next short lesson	30 s suggestion
COMPLETE	Success or player ends	Return home	None

9.2 Agility course contract

```
{
  "courseId": "crs_01J...", "version": 7, "siteId": "site_01J...",
  "anchorId": "vps-anchor-id", "mode": "RANKED", "ruleset": "AGILITY_V1",
  "objects": [
    {"order": 0, "type": "START_GATE", "pose": {"p": [0,0,0], "q": [0,0,0,1]}},
    {"order": 1, "type": "JUMP_LOW", "pose": {"p": [3,0,1], "q": [0,0.2,0,0.98]}},
    {"order": 2, "type": "FINISH_GATE", "pose": {"p": [7,0,2], "q": [0,0,0,1]}}
  ],
  "safetyRevision": 4, "contentDigest": "sha256:..."
}
```

- A ranked course SHALL contain exactly one start gate, one finish gate, and 1-20 ordered obstacles.
- The solver SHALL prove a continuous navigation corridor at least 0.8 m wide and 1.5 m high, with no hazard tag intersection and no obstacle closer than 1.5 m to the camera start volume.
- Gate crossing SHALL use swept-volume intersection between the pet root capsule and ordered gate plane, not frame-point containment.
- The authoritative timer SHALL use server-issued startEpochMs and monotonic client deltas. Client wall-clock values are never trusted for ranking.
- A tracking loss > 3.0 s, course digest mismatch, impossible velocity, out-of-order gate, app backgrounding, or safety intervention SHALL mark the run UNRANKED. The player may still finish for practice credit.
- A course version SHALL be immutable after publication. Any object, transform, rule, site, or safety change creates version +1 and invalidates cached ranking eligibility for the prior draft.

10. Backend services and data

10.1 Technology baseline

Concern	Selected technology	Normative rule
Service runtime	Node.js LTS + TypeScript strict	No implicit any; runtime input validation; lockfile pinned
HTTP	REST JSON over TLS 1.3	OpenAPI 3.1 is contract source; generated clients only
Realtime	WebSocket over TLS	Sequence-numbered snapshots/deltas; reconnect token; bounded replay
Primary data	PostgreSQL + PostGIS	UTC timestamptz; migrations forward-only; row ownership enforced
Cache/ephemeral	Redis	No durable authority; TTL required; fail safe on loss
Jobs	Redis-backed queue	At-least-once; idempotent handlers; dead-letter queue
Object storage	Backblaze B2 S3-compatible + CDN	Private buckets; immutable digest keys; short-lived signed fetch
Telemetry	OpenTelemetry	Trace ID propagated; PII filter before export
Infrastructure	OpenTofu/Terraform	No manual production resources; plan reviewed

10.2 Core tables

Table	Key fields	Invariant
users	user_id, auth_subject, birth_band, status, locale, created_at	Identity reference; no password hashes if external IdP
pet_assets	pet_asset_id, issuer, source_id, version, species, digest, manifest_json, key_id, status	Immutable published versions; status can revoke
pet_entitlements	user_id, pet_asset_id, version, status, granted_at, revoked_at	Exact asset ownership
pets	pet_id, user_id, pet_asset_id, display_name, personality_seed, state_revision	Player companion instance
pet_state	pet_id, bond, energy, training_json, updated_at	Server-authoritative durable state
pet_memories	memory_id, pet_id, fact_type, fact_json, consent, status, created_at	Allowlisted AI/game memory
sites	site_id, provider_site_id, geography, access, safety_status, version	Spatial discovery and site access
courses	course_id, site_id, current_version, owner_id, status	Course identity
course_versions	course_id, version, anchor_id, content_json, digest, safety_revision	Immutable published payload
course_runs	run_id, course_id, version, user_id, pet_id, result, time_ms, proof_digest	Ranking evidence and audit
inventory_ledger	entry_id, user_id, item_id, quantity_delta, reason, idempotency_key	Append-only value ledger
outbox_events	event_id, aggregate_type, aggregate_id, event_type, payload, published_at	Transactional event delivery
audit_log	audit_id, actor, action, target, before_hash, after_hash, occurred_at	Admin and security audit

- All public IDs SHALL be opaque, globally unique, prefixed identifiers. Sequential database IDs SHALL NOT cross the API boundary.
- Every mutable aggregate SHALL contain a bigint revision incremented in the same transaction as the change. Mutations MAY require If-Match revision.
- Inventory SHALL be an append-only ledger. Current balance is derived and may be cached; negative balance constraints are enforced transactionally.
- Soft deletion SHALL use status and deleted_at for user-facing entities. Security revocation SHALL take effect immediately through status checks and cache invalidation.
- Database backups SHALL be encrypted, daily, retained 35 days, and restore-tested quarterly. Target RPO 15 minutes and RTO 2 hours for primary game data.

11. API contract

OpenAPI 3.1 is the single source of truth. Requests and responses use application/json UTF-8, RFC 3339 UTC timestamps, lowerCamelCase fields, and opaque string IDs. Unknown response fields must be ignored by clients; unknown request fields are rejected for security-sensitive endpoints.

Endpoint	Auth/input	Response	Concurrency/retry
POST /v1/auth/exchange	External identity token	Access/refresh token pair	None
GET /v1/pets	Bearer	Owned pet instances and catalog state	None
POST /v1/pets	Bearer + Idempotency-Key	Create pet from active entitlement	Idempotent 24 h
GET /v1/assets/{id}/manifest	Bearer	Signed manifest and short fetch descriptor	ETag by digest
POST /v1/integrations/pawsome/webhook	HMAC + timestamp	Accept asset-ready event	eventId unique
POST /v1/training/sessions	Bearer + key	Start deterministic lesson	Idempotent
POST /v1/training/sessions/{id}/events	Bearer + sequence	Append cue/action/reward	Sequence exact
GET /v1/world/nearby	Bearer + geo scope	Sites and discovery cards	Cache 30 s
GET /v1/sites/{id}/course-versions/{v}	Bearer	Immutable course payload	ETag by digest
POST /v1/course-runs	Bearer + key	Create run and server start token	Idempotent
POST /v1/course-runs/{id}/finish	Bearer + proof	Ranked/unranked result	One terminal write
POST /v1/ai/intents	Bearer + context revision	Validated high-level intent	Rate-limited; no retry storm
DELETE /v1/pets/{id}/memories/{mid}	Bearer	Tombstone memory	Idempotent
POST /v1/reports	Bearer + key	Safety/moderation report	Idempotent

11.1 Error envelope

```
{
  "error": {
    "code": "ASSET_SIGNATURE_INVALID",
    "message": "This pet could not be verified.",
    "requestId": "req_01J...",
    "retryable": false,
    "details": {"petAssetId": "asset_01J..."}
  }
}
```

Code	HTTP	Retry	Client action
AUTH_REQUIRED	401	No	Sign in again
FORBIDDEN	403	No	Generic access message
REVISION_CONFLICT	409	Yes after refresh	Reload state
IDEMPOTENCY_CONFLICT	409	No	Do not repeat with changed body
ASSET_NOT_ENTITLED	403	No	Reconnect correct Pawsome3D account
ASSET_SIGNATURE_INVALID	422	No	Disable asset; contact support
ASSET_REVOKED	410	No	Fallback to preset/select another
ANCHOR_NOT_TRACKED	409	Yes	Relocalize
COURSE_VERSION_STALE	409	Yes after fetch	Download current version
RUN_NOT_RANKED	200 result	No	Finish as practice
RATE_LIMITED	429	Yes	Honor Retry-After

Code	HTTP	Retry	Client action
DEPENDENCY_UNAVAILABLE	503	Yes	Exponential backoff + jitter

11.2 Idempotency and retries

- Every POST that can create value or a durable entity SHALL accept Idempotency-Key. The key scope is authenticated subject + method + route; retention is 24 hours minimum.
- A repeated key with the same canonical body returns the original status and body. A repeated key with a different body returns IDEMPOTENCY_CONFLICT.
- Automatic retries are allowed only for connection failure, 408, 429, and 5xx when the operation is idempotent. Backoff is 0.5 s, 1 s, 2 s, 4 s plus 0-250 ms jitter, maximum 4 retries.
- The client SHALL never retry ASSET_SIGNATURE_INVALID, validation 4xx, or payment/inventory terminal results without a fresh user action.

12. Realtime shared AR

- All participants SHALL localize to the same siteId, courseId, courseVersion, and anchorId before joining a shared AR room.
- Network poses SHALL be anchor-local position and normalized quaternion, never device-local world coordinates or precise player GPS.
- Pet transforms SHALL send at 15 Hz with uint32 sequence, server timestamp, position quantized to 1 mm, rotation compressed, locomotion state, and animation phase. Clients interpolate with a 100 ms buffer.
- The server SHALL reject speed > 8 m/s, acceleration > 20 m/s2, teleport > 2 m without a signed reset event, and sequence regression.
- Shared sessions SHALL cap at 8 players for v1. Proximity voice is out of scope; approved emotes and preset phrases are supported.
- When disconnected, remote pets fade after 2 s and disappear after 10 s. Local play continues. Rejoin requires a new room token and anchor eligibility check.

13. Security and privacy

13.1 Trust boundaries

- The mobile device, network, pet payload, course author input, AI output, and all webhooks are untrusted inputs.
- TLS 1.3 is required for public production endpoints. Certificate validation SHALL NOT be disabled. Secrets SHALL be stored in the deployment secret manager and rotated at least every 90 days or immediately after suspected exposure.
- Access tokens SHALL live <= 15 minutes; refresh tokens SHALL rotate on use and be stored in Keychain/Keystore. Tokens SHALL be audience- and environment-scoped.

- Admin access SHALL require SSO, hardware-key MFA, role-based authorization, device posture where available, and immutable audit logging.
- Webhook signatures SHALL include timestamp and event ID. Events older than 5 minutes or previously consumed are rejected.
- Object storage SHALL be private. Publicly guessable object keys, bucket listing, and permanent asset URLs are forbidden.
- The AI provider egress path SHALL have a strict allowlist and structured request logger that redacts player text and identifiers by default.

13.2 Data classification and retention

Data	Class	Default retention	Rule
Account and entitlement	Confidential	Account life + legal window	Required to provide service; deletion workflow applies
Precise current location	Sensitive	In memory; <= 24 h security logs if flagged	No raw history by default
VPS provider user ID	Sensitive pseudonymous	Provider/project policy	Disclose and rotate where supported
Camera frames	Restricted	No GibiWorld storage	Processed by AR provider; cloud VPS disclosure required
Voice audio	Restricted	No storage by default	On-device recognition preferred; opt-in required
Pet memory	Confidential	Until user deletes or account closes	Allowlisted facts only; export/delete supported
Course run proof	Confidential	90 d standard; 365 d disputed	Minimize location; hash raw evidence
Telemetry	Internal	30 d raw, 13 mo aggregate	No precise location, tokens, asset URLs, or camera data
Safety report	Restricted	As policy/legal requires	Access logged; separate storage

13.3 Player safety gates

- AR interactions SHALL pause when speed exceeds 4.5 m/s for 10 seconds. Map navigation may continue in passenger-safe mode, but no collection, placement, training, or course actions are enabled.
- Published course geometry SHALL be inside an approved site polygon, outside exclusion polygons, and at least 10 m from mapped road centerlines and rail corridors. Venue operators may require stricter setbacks.
- On-device validation SHALL require a clear camera start volume, traversable floor, acceptable slope, lighting confidence, and no currently detected person/vehicle intersection.
- A player can report an unsafe or inaccessible site in two taps. A threshold breach or credible moderator review disables discovery immediately without deleting evidence.
- The app SHALL remind players that computer vision and map data are imperfect and surroundings outrank game prompts.

14. Accessibility, localization, and UX rules

- Every critical AR state SHALL have text, shape/icon, color, audio/caption, and haptic mappings where the device supports them.
- Tap targets SHALL be at least 44 x 44 points on iOS-equivalent layouts and 48 x 48 dp on Android-equivalent layouts.
- Voice is optional. Every voice cue SHALL have a tap or gesture alternative; every sound-only event SHALL have caption/visual equivalent.
- Reduced motion SHALL disable screen shake, forced camera motion, rapid scale changes, and high-velocity particles while preserving gameplay timing.
- All player-visible strings SHALL use localization keys. No concatenated sentences, hardcoded pluralization, or text embedded in textures.
- World distances SHALL be stored in meters and formatted for locale. Identifiers, hashes, timestamps, and error codes remain invariant.

15. Observability and service levels

SLI	Target	Measurement window	Response
Core API availability	>= 99.9%	28-day rolling	Page on 5-minute burn-rate breach
Asset manifest API p95	<= 300 ms	5-minute and 24-hour	Investigate cache/DB; fail closed
World nearby API p95	<= 500 ms	5-minute and 24-hour	Serve bounded stale cache <= 5 min
AI intent p95	<= 2.5 s	15-minute	Use local fallback; do not block play
Realtime room availability	>= 99.5%	28-day	Allow solo fallback
Crash-free sessions	>= 99.7%	Per release/device tier	Halt rollout on breach
Ranked proof acceptance	>= 99% for eligible runs	Daily	Disable affected ruleset if systemic
Asset integrity failures	0 expected	Immediate	Security alert and asset quarantine

- Every request SHALL include or receive requestId. Trace context SHALL propagate through gateway, service, database, queue, and provider adapters.
- Structured logs SHALL name event, severity, environment, service, version, requestId, user pseudonym when required, and error code. Free-form stack traces are restricted to protected logs.
- Metrics SHALL include anchor state duration, relocalization count, AR session startup, frame time, thermal state, asset validation outcomes, AI fallback rate, course invalidation reasons, and safety pauses.
- Release dashboards SHALL segment by client version, OS, device tier, country/region at coarse level, network type, and feature flag without exposing precise location.

16. CI/CD and environments

Environment	Data	Providers	Promotion rule
local	Synthetic only	Playback/spatial mocks; local services	Developer tests pass
dev	Synthetic/test accounts	NSDK development project; AI sandbox	Main branch build
staging	Production-shaped synthetic	Separate spatial, storage, push, AI keys	Release candidate + QA sign-off
production	Real player data	Production projects and keys	Signed change ticket + canary

- Client builds SHALL be reproducible from a signed Git commit, Unity editor revision, packages-lock.json hash, Addressables/content catalog hash, and build configuration manifest.
- CI SHALL run formatting, compile, static analysis, secret scan, dependency audit, EditMode tests, PlayMode tests, asset validator fixtures, OpenAPI compatibility, database migration test, and mobile smoke builds.
- Production deployment SHALL be rolling or blue/green with database migrations separated into expand, migrate, contract phases. A release shall not require simultaneous client and server cutover.
- Feature flags SHALL have owner, purpose, default, expiry date, kill switch, and environment scope. Expired flags fail CI after a 14-day grace period.
- Mobile rollout SHALL use internal, 1%, 5%, 20%, 50%, 100% stages with at least 24 hours at 5% and later stages unless an emergency exception is approved.

17. Test and acceptance gates

Requirement	Normative behavior	Acceptance test
GW-AR-001	Exactly one ARSession and XR Origin are active in ARWorld.	Automated scene validation
GW-AR-002	Ranked content is solid and score-enabled only while target anchor state is tracked.	Recorded provider state transition test
GW-AR-003	Tracking loss pauses ranked time within 250 ms.	Device instrumentation
GW-AR-004	A local anchor never reaches persistence endpoint.	Unit + API contract test
GW-AR-005	Anchor-local quaternion is normalized within 1e-4.	Property test
GW-AR-006	Hazard semantic regions reject destinations and obstacles.	Playback fixtures
GW-AR-007	Slope and clearance gates match published thresholds.	Geometry test suite
GW-AR-008	Relocalization recovery does not move course child poses > 2 cm.	Mapped-site device test
GW-ASSET-001	Unknown issuer, key, schema, species, or rig rejects before render.	Tampered fixture matrix
GW-ASSET-002	Digest mismatch deletes temporary bytes and does not retry URL.	Integration test
GW-ASSET-003	Expired or revoked entitlement blocks pet instantiation.	Auth integration test
GW-ASSET-004	External URI in GLB rejects.	Malicious GLB fixture
GW-ASSET-005	Geometry, texture, bone, material, and clip limits are enforced server and client.	Boundary fixtures

Requirement	Normative behavior	Acceptance test
GW-ASSET-006	Asset cannot supply shader, script, or animation event execution.	Static and runtime inspection
GW-ASSET-007	Cache promotion is atomic and keyed by SHA-256 digest.	Crash-injection test
GW-ASSET-008	Preset and Pawsome3D assets use the same runtime verifier.	End-to-end test
GW-GAME-001	Safety override interrupts all lower-priority behavior within one 10 Hz tick.	Simulation test
GW-GAME-002	Locomotion distance is deterministic across 30, 60, and 120 fps render rates.	Frame-rate variance test
GW-GAME-003	Ranked gate crossing uses swept volume and exact order.	Physics test
GW-GAME-004	App backgrounding makes active ranked run unranked.	Lifecycle test
GW-GAME-005	Offline mode supports placement, direct cues, training, fetch, and practice course.	Airplane-mode playthrough
GW-GAME-006	No absence duration decreases bond or inventory.	Clock-jump and 90-day absence test
GW-GAME-007	Course versions are immutable after publication.	Database constraint test
GW-GAME-008	Tracking degradation lets player finish as practice without leaderboard write.	End-to-end test
GW-AI-001	AI output validates against fixed JSON schema and intent enum.	Fuzzed provider output
GW-AI-002	AI cannot directly select animations, coordinates, physics, value, score, or moderation.	Schema negative tests
GW-AI-003	Expired, stale-revision, or wrong-target intent is rejected.	Unit test
GW-AI-004	AI timeout silently invokes local fallback.	Dependency fault injection
GW-AI-005	Provider request excludes precise location, camera frames, tokens, contacts, and raw voice.	Egress assertion test
GW-AI-006	Deleted memory is absent from AI context within 24 hours.	Deletion workflow test
GW-API-001	OpenAPI 3.1 validation rejects invalid security-sensitive fields.	Contract test
GW-API-002	Idempotency returns original response for same key/body and conflict for changed body.	Concurrency test
GW-API-003	Revision conflict returns 409 without partial mutation.	Transactional test
GW-API-004	Client retries only approved status/transport failures with bounded jitter.	Network proxy test
GW-API-005	Tokens and signed URLs never appear in logs or crash payloads.	Redaction canary
GW-API-006	Inventory ledger cannot produce negative balance under concurrency.	Load/property test
GW-SEC-001	Production public endpoints require TLS 1.3 and valid certificates.	External scanner
GW-SEC-002	Refresh token rotates on use and old token reuse revokes family.	Security integration test
GW-SEC-003	Webhook timestamp window and event replay protection are enforced.	Replay test
GW-SEC-004	Production buckets are private and listing is disabled.	Cloud policy test
GW-SEC-005	Admin requires SSO, hardware MFA, RBAC, and audited mutations.	Access review
GW-SEC-006	Telemetry filter removes precise coordinates and credentials before export.	PII canary

17.1 Device test matrix

Tier	Minimum representative set	Required scenarios
iOS Tier A	Recent LiDAR Pro device	Depth/mesh, VPS2 tracked, long course, thermal, low light
iOS Tier B	A12-class non-LiDAR device	NSDK depth, local anchor, asset load, memory pressure
Android Tier A	Recent flagship ARCore device	VPS2, meshing, occlusion, 60 fps, network handoff
Android Tier B	Midrange ARCore device, 6 GB RAM	30/60 quality switch, thermal, camera resume
Android Tier C	Minimum supported ARCore device	30 fps fallback, no depth path, low storage

18. Release phases

Phase	Exit scope	Availability
P0 - Vertical slice	One preset dog + one Pawsome3D pet; local placement; sit/fetch; one local course; offline fallback	Internal devices only
P1 - Closed alpha	Signed asset pipeline; account/entitlement; VPS2 test sites; telemetry; safety reports	Denver + Philadelphia curated sites
P2 - Beta	Course authoring/publishing; ranked proof; shared sessions; accessibility; AI intent gateway	Invite rollout, 13+
P3 - Launch	Operational SLOs; moderation; incident runbooks; 1%-100% rollout; nationwide local play	VPS content only at approved sites
P4 - Expansion	Additional domestic species, venue tools, more shared experiences, optional headset experiments	Separate ADR and performance profile

19. Required implementation artifacts

- Unity repository with Assembly Definition dependency checks, pinned ProjectVersion.txt, manifest.json, packages-lock.json, automated scene validator, and device quality profiles.
- OpenAPI 3.1 specification, generated TypeScript server types, generated C# client DTOs, JSON Schema for manifests/courses/AI intents, and backward-compatibility CI.
- Pawsome3D adapter contract and deployment mapping for real endpoint/env names; HMAC rotation runbook; integration fixtures with no production credentials.
- Blender/GLB validator container with deterministic version manifest, malicious/boundary fixtures, signing service client, provenance output, and quarantine policy.
- Database migrations, ERD, retention jobs, backup/restore runbook, inventory concurrency tests, and PostGIS spatial index benchmark.
- Threat model, privacy data-flow map, vendor data-processing inventory, age-gate policy, app-store privacy declarations, accessibility test plan, and safety-site review checklist.
- Dashboards and alerts for all SLOs, asset integrity, ranked invalidations, AI fallbacks, tracking state, frame time, thermal events, and rollout health.

- Release checklist binding each GW-* requirement to an automated result or signed manual evidence. A missing result is a failed release gate.

20. Architecture decision records required before code freeze

ADR	Decision	Default in this spec
ADR-001	Production authentication provider and account-linking model	External OIDC; backend token exchange
ADR-002	Exact Render topology vs later container platform	Stateless Render services for MVP; provider-neutral IaC
ADR-003	Realtime transport implementation	WebSocket v1; anchor-local protocol
ADR-004	AI model provider and regional data controls	Provider adapter; no provider SDK in client
ADR-005	Launch species allowlist and skeleton profiles	Dog/cat primary; small domestic mammals gated
ADR-006	Venue/site publication authority and map data source	Curated operator workflow; fail-closed exclusions
ADR-007	Commerce, subscription, and item economy	No paid random rewards; ledger ready but commerce out of scope

Appendix A. Source baseline

Source	URL	Why used
Niantic Spatial SDK setup, release 4.1.0	https://www.nianticspatial.com/docs/nsdk/setup/?platform=unity	Pins Unity 6000.0.74f1 and recommends AR Foundation 6.4.1.
Niantic Spatial VPS2	https://www.nianticspatial.com/docs/nsdk/features/vps2/?platform=unity	Defines coarse vs precise localization and separate anchor tracking state.
Niantic Spatial meshing	https://nianticspatial.com/docs/nsdk/how-to/ar/meshing/adding_meshing/?platform=unity	Defines ARMeshManager integration, semantic filtering, LiDAR/non-LiDAR behavior.
Unity AR development overview	https://docs.unity3d.com/6000.6/Documentation/Manual/AROverview.html	AR Foundation scene and provider architecture.
Unity glTFast 6.16.1	https://docs.unity3d.com/Packages/com.unity.cloud.gltfast%406.16/manual/index.html	Runtime glTF 2.0 import and compression support.
Unity Addressables 1.22.3	https://docs.unity3d.com/Packages/com.unity.addressables%401.22/manual/index.html	Remote Unity-authored content and dependency management.

Version note: vendor packages and platform rules change. This specification intentionally pins a tested baseline. Upgrades require an ADR, dependency diff, recorded AR playback regression, asset fixture validation, device matrix pass, and staged rollout.

Appendix B. Glossary

Term	Definition
Anchor-local	A pose stored relative to one accepted spatial anchor rather than a device session origin.

Term	Definition
Entitlement	Server authority that a specific user may use a specific immutable pet asset version.
GLB	Single-file binary packaging of glTF 2.0 model, textures, skeleton, and animation data.
Local play	Unranked AR that uses a device-local plane or mesh anchor and does not persist world placement.
NSDK	Niantic Spatial SDK.
Pawsome3D	The approved external pet authoring and source-ownership platform.
Precise	VPS2 device localization resolved against a processed Site map.
Tracked	Per-anchor state indicating reliable map-relative content placement.
VPS2	Niantic Spatial Visual Positioning System 2.